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Research Proposals on

Diagnostic Gaps in Cardiovascular Diseases: Exploring Unmet Needs in the Early Detection of Acute and Intermittent Cardiac Events

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Cardiovascular diseases (CVDs) are a major health problem worldwide, and finding them early can change how they are treated [1]. But having a test does not mean the test will catch a problem when it happens. Some heart events are sudden, come and go, or change over time, while a test only watches the patient for a short period. A resting electrocardiogram (ECG) shows the electrical activity of the heart, but one recording gives only a short snapshot. An abnormal rhythm that comes and goes may not appear during that time. This is a common problem with paroxysmal arrhythmias (irregular heartbeats that start and stop suddenly). Whether they are caught depends partly on how often they happen and how long the heart is recorded [2], [3].

A similar problem happens in acute coronary syndromes (ACS), such as a heart attack in progress. A first ECG that looks normal does not rule out a problem that is still developing, so guidelines suggest repeating the ECG and checking cardiac biomarkers (blood tests) when appropriate [4]–[6]. This shows that a single measurement at one moment may not be enough in some situations.

To catch events that come and go, monitoring has gradually become longer. Options include repeated ECGs, Holter monitors, patient-activated event recorders, external loop recorders, mobile cardiac telemetry (MCT), implantable loop recorders (ILRs), and wearable ECG devices [2], [3], [7]–[10]. More recently, wearable devices that use ECG, photoplethysmography (PPG, which measures the pulse using light), and other sensors have made increasingly continuous monitoring possible outside the hospital [3], [11], [12]. However, longer monitoring brings new problems. Recording for longer does not guarantee that the data will be good enough to use. Motion artefacts (noise caused by movement), poor contact between the sensor and the skin, signal noise, patients not wearing the device, short battery life, and the effort of reviewing large amounts of data can all reduce the value of long-term monitoring [11]–[15]. Recent reviews also point to real-world validation, signal reliability, ease of use, and fitting into clinical workflow as ongoing challenges for wearable heart monitoring [14]–[16].

Long-term monitoring already exists. The difficulty is getting four things at the same time: long coverage in time, a reliable signal that is good enough for diagnosis, enough physiological information, and acceptable use in daily life.

Existing Literature and Current Monitoring Approaches

Different monitoring methods solve different parts of the problem.A resting ECG gives high-quality electrical information, but it is short because it is usually taken during one clinic visit. Serial ECG (repeated ECGs) adds more time points and is useful when a problem may change over time, such as in suspected ACS [4]–[6]. Still, it depends on the patient being tested at chosen times.

Holter monitors record the ECG while the patient goes about daily life, and they are useful when symptoms or arrhythmias happen fairly often. Event recorders and external loop recorders watch for longer and can record when the patient has symptoms or when the device detects an abnormal rhythm. Mobile cardiac telemetry can monitor almost continuously and send data automatically in selected cases [2], [3], [7].

Implantable loop recorders cover a much longer time and are especially useful for rare events. However, they need a procedure to implant them, and only certain patients have a clear reason to receive one, so they cannot be the answer for everyone [9], [10].

Wearable ECG and PPG devices allow monitoring during everyday activities. They are portable and increasingly use artificial intelligence (AI), which opens the way to longer monitoring and automatic detection [3], [11], [12]. Recent reviews report progress in wearable ECG, soft materials, AI-based interpretation, and the use of several sensor types together. Even so, these devices are held back by signal quality, motion artefacts, difficulty explaining how AI reaches its results (interpretability), regulation, and limited testing in varied real-world settings [14]–[16].

The literature therefore shows trade-offs across several areas:

Monitoring dimension	Existing approaches	Remaining concern
Temporal coverage	Holter, MCT, ILR, wearables	Longer monitoring increases exposure to real-world artefacts
Diagnostic-quality signal	Clinical ECG, selected ambulatory ECG	Signal quality may deteriorate during prolonged everyday use
Physiological information	ECG, PPG, additional wearable sensors	Different modalities provide different types and levels of information
Continuity	Continuous or semi-continuous wearable monitoring	Data gaps and poor-quality periods may occur
Usability	Wearable and non-invasive systems	Comfort, adherence, charging, placement, and user burden remain relevant
Data interpretation	Automated detection and AI	Large data volumes, robustness, interpretability, and clinical validation remain challenges

Recent work on AI-based wearable heart monitoring shows that systems with several sensor types can combine ECG with PPG, acceleration, blood pressure, skin temperature, and other signals. However, many studies are still short, based in hospitals, or not properly tested in unsupervised real-life conditions [15]. This suggests that the next step may be to combine sensing, signal-quality checking, adaptive recording, and data interpretation in one working system, rather than to build one more sensor.

Gap Validation

Several possible research gaps can be seen in the literature.

First, a gap in monitoring length is already mostly covered, because long-duration monitoring tools exist. So the problem cannot be described simply as “current monitoring is too short.” Second, there is still a trade-off between monitoring length and the richness of the information. Tools that can watch for a long time may not always give the same diagnostic-quality information as a clinical ECG. This matters most when the person is moving or doing normal daily activities.Third, there is a gap between getting a signal and taking action. Finding an abnormal signal does not automatically give information that a doctor can act on. But this problem is wider than sensing technology. It also involves clinical workflow, interpretation, and decision-making.

The most relevant gap is therefore an integration gap across several needs. Current technologies have greatly extended monitoring time, and they have started to improve the information collected, automatic detection, and ease of use. But the evidence does not yet clearly show that long coverage, reliable diagnostic-quality signal, enough physiological information, and acceptable real-world use can all be kept together.

This is a technology need that is still not fully solved for monitoring heart problems that come and go or change over time.

Research Question

How can existing cardiac monitoring technologies be combined or adapted so that they achieve, at the same time, long monitoring time, reliable diagnostic-quality signal, enough physiological information, and acceptable real-world use, for detecting heart events that come and go or change over time?

Research Aim

This study aims to explore diagnostic gaps in cardiovascular diseases, focusing on how time limits affect the detection of sudden and intermittent cardiac events, and to develop a conceptual framework for a new monitoring approach that addresses the gaps found.

Proposed Innovative Solution

Based on the integration gap, this study proposes an Adaptive Multimodal Cardiac Monitoring Framework. The framework does not introduce a completely new stand-alone device. It combines several existing ideas into one adaptive monitoring system. The first part is long-term cardiac monitoring, to cover more time than a short clinical recording. The second part is continuous signal-quality checking, so the system can tell useful diagnostic signals apart from periods affected by motion artefacts, poor sensor contact, or other noise.

The third part is adaptive recording. Instead of treating all monitoring time in the same way, the system could change how it records when it detects a suspicious change or an important clinical event. It could record with higher quality or more detail around important events, and use lighter monitoring during stable periods. The fourth part is multimodal sensing. ECG could be combined with other signals, such as PPG or movement measurements, to give more context. Current literature already shows growing interest in multimodal wearable heart monitoring, but real-world validation is still limited [14]–[16].

The last part is AI-assisted event prioritization, which would find possibly important segments in large amounts of long-term data. It would not replace a doctor's diagnosis. Its aim is to reduce the data that must be reviewed by hand and to point clinicians to the events that matter. Recent reviews and consensus statements describe substantial progress in AI-based ECG analysis, but also limits in validation, interpretability, bias, and real-world use [15]–[17].

So the novelty of this proposal is mainly in combining and adapting existing parts. It does not claim that each part is new.

Significance and Expected Contribution

This study should give a clearer understanding of why longer cardiac monitoring does not always give continuous, diagnostic-quality information. The framework could be a starting point for future monitoring systems that balance monitoring time with signal quality, physiological information, and real-world usability.
In particular, adaptive recording and multimodal sensing may reduce the information lost during long monitoring, and AI-assisted prioritization may help deal with the growing amount of long-term physiological data.

More broadly, the study may offer a way to judge future monitoring technologies, not only by how long they can record, but also by whether the recorded data are useful for diagnosis, informative enough, and practical to use in real life.

Significance and Expected Contribution

This study focuses on technology gaps in the monitoring and early detection of acute, intermittent, and evolving cardiac events, especially ECG-based and wearable monitoring technologies.It will examine existing monitoring methods and newer approaches that use wearable sensing, multimodal physiological monitoring, signal-quality assessment, and AI-assisted analysis. It will not set new clinical diagnostic criteria, and it does not claim that AI can replace clinician interpretation.

The proposed framework will stay conceptual in this study. Prototype development, large-scale clinical validation, regulatory assessment, and comparison with established clinical devices are suggested as later research.

Reference

[1]World Health Organization, "Cardiovascular diseases (CVDs)," Fact sheet, Jul. 31, 2025. [Online]. Available:https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds). Accessed: Sep. 20, 2026.

[2]J. S. Steinberg et al., "2017 ISHNE-HRS expert consensus statement on ambulatory ECG and external cardiac monitoring/telemetry," Heart Rhythm, Jul. 2017
Available: https://www.heartrhythmjournal.com/article/S1547-5271(17)30415-0/fulltext

[3]T. Ikeda et al., "2025 Japanese Heart Rhythm Society/Japanese Circulation Society consensus statement on the appropriate use of ambulatory and wearable electrocardiographs," J. Arrhythmia, Art. no. e70059, 2025,Available: https://onlinelibrary.wiley.com/doi/10.1002/joa3.70059

[4] S. V. Rao et al., "2025 ACC/AHA/ACEP/NAEMSP/SCAI guideline for the management of patients with acute coronary syndromes: A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines," J. Am. Coll. Cardiol., vol. 85, no. 22, pp. 2135–2237, Jun. 2025, Available: https://www.jacc.org/doi/10.1016/j.jacc.2024.11.009

[5] M. Gulati et al., "2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain," Circulation, vol. 144, no. 22, pp. e368–e454, Nov. 2021, Available: https://www.ahajournals.org/doi/10.1161/CIR.0000000000001029

[6] R. F. Riley et al., "Diagnostic time course, treatment, and in-hospital outcomes for patients with ST-segment elevation myocardial infarction presenting with nondiagnostic initial electrocardiogram: A report from the American Heart Association Mission: Lifeline program," Am. Heart J., vol. 165, no. 1, pp. 50–56, Jan. 2013, Available: https://doi.org/10.1016/j.ahj.2012.10.027

[7] S. A. Rothman et al., "The diagnosis of cardiac arrhythmias: A prospective multi-center randomized study comparing mobile cardiac outpatient telemetry versus standard loop event monitoring," J. Cardiovasc. Electrophysiol., vol. 18, no. 3, pp. 241–247, 2007.

[8] S. R. Steinhubl et al., "Effect of a home-based wearable continuous ECG monitoring patch on detection of undiagnosed atrial fibrillation: The mSToPS randomized clinical trial," JAMA, vol. 320, no. 2, pp. 146–155, Jul. 2018, Available: https://doi.org/10.1001/jama.2018.8102

[9] V. Vilcant, O. Kousa, and O. Hai, "Implantable loop recorder," in StatPearls. Treasure Island, FL, USA: StatPearls Publishing, 2023. [Online]. Available: https://www.ncbi.nlm.nih.gov/books/NBK470398/

[10] M. Janmohamadi et al., "Value of implantable loop recorders in detecting atrial fibrillation for stroke prevention: A systematic review and meta-analysis," Open Heart, vol. 12, no. 2, Art. no. e003456, Sep. 2025, Available: https://doi.org/10.1136/openhrt-2025-003456

[11] E. S. Dahiya, A. M. Kalra, A. Lowe, and G. Anand, "Wearable technology for monitoring electrocardiograms (ECGs) in adults: A scoping review," Sensors, vol. 24, no. 4, Art. no. 1318, Feb. 2024, Available: https://www.mdpi.com/1424-8220/24/4/1318

Reference

[13] D. K. Saggu et al., "Feasibility of using chest strap and dry electrode system for longer term cardiac arrhythmia monitoring: Results from a pilot observational study," Indian Pacing Electrophysiol. J., vol. 24, no. 5, pp. 282–290, 2024, Available: https://doi.org/10.1016/j.ipej.2024.08.003

[14] C. Pedrini Schuch et al., "Applications of smart textiles for ambulatory electrocardiogram monitoring: Scoping review of the literature," JMIR Cardio, vol. 10, Art. no. e74261, 2026, Available: https://doi.org/10.2196/74261

[15] A. Abedi et al., "AI-driven real-time monitoring of cardiovascular conditions with wearable devices: Scoping review," JMIR mHealth uHealth, vol. 13, Art. no. e73846, Nov. 2025, Available: https://doi.org/10.2196/73846

[16] B. Khan, R. T. Khalid, M. H. Masrur, M. Awais, B. Khan, and B. L. Khoo, "Next-generation wearable ECG systems: Soft materials, AI integration, and personalized healthcare applications," Chem. Eng. J., vol. 525, Art. no. 170117, Dec. 2025, Available: https://doi.org/10.1016/j.cej.2025.170117

[17] E. Svennberg et al., "State of the art of artificial intelligence in clinical electrophysiology in 2025: A scientific statement of the European Heart Rhythm Association (EHRA) of the ESC, the Heart Rhythm Society (HRS), and the ESC Working Group on E-Cardiology," EP Europace, vol. 27, no. 5, Art. no. euaf071, May 2025, Available: https://doi.org/10.1093/europace/euaf071